Self-Dual BKZ on NTRU: Dimension-Onset, Fatigue, and a Cross-Engine Reference-Free DSD Observable

Brendan Chambers-Bourgeois

Independent Researcher, Melbourne, Australia

Abstract

This technical report extends the self-dual BKZ (SD-BKZ) reduction benchmark of the companion paper from LWE-Kannan lattices to *NTRU*, and from a single enumeration engine (fplll) to a second, sieving engine (G6K). We report three empirical results. First, on circulant NTRU lattices the SD-BKZ advantage exhibits a sharp *dimension-onset*: tied with BKZ for small n, a high-variance spike around $n \approx 71$–73 (mean advantage up to $+9.7$ nats), then a tight $+1.5$-nat plateau for $n \geq 79$. Second, the defensible NTRU observable is *reference-free*: signed $d(\mathrm{LN})$ is a reference artifact on NTRU (no canonical fixed point), but the BKZ-vs-SD-BKZ profile divergence / dense-sublattice-discovery (DSD) onset is robust, and the SD-vs-BKZ DSD-onset *gap grows with dimension* ($0\% \to 27\%$ over $n = 67$–113) as q crosses the fatigue point. Third, the cross-engine study confirms the companion paper's thesis from the other side—the Root Hermite Factor is blind to the SD-BKZ difference while $d(\mathrm{LN})$/DSD sees it—now on a sieve oracle, and finds the effect to be regime-dependent. We also re-validate, on a new modulus family, an fplll GSO numerical fix. All experiments are reproducible via a containerised, determinism-gated benchmark.

Keywords: lattice reduction, BKZ, SD-BKZ, NTRU, overstretched, fatigue, dense sublattice discovery, Rankin profile, sieving, G6K

1 Introduction

The companion paper [CB26] showed, on LWE-Kannan lattices, that the Root Hermite Factor (RHF) is structurally blind to differences between BKZ and self-dual BKZ [MW16] that the Li–Nguyen Rankin profile distance $d(\mathrm{LN})$ [LN20] makes visible. This report asks two follow-up questions a data paper is well suited to answer: *(i) does the SD-BKZ phenomenology generalise beyond LWE to NTRU?* and *(ii) does it reproduce on a fundamentally different reduction oracle—sieving (G6K) rather than enumeration?*

Our contributions are primarily empirical and infrastructural:

- A characterisation of the SD-BKZ advantage on circulant NTRU lattices [DvW21], including a sharp dimension-onset transition (Section 4).
- Evidence that the appropriate NTRU observable is reference-free, and a *dense-sublattice-discovery (DSD) onset* whose SD-vs-BKZ gap grows with dimension as q approaches the fatigue point (Section 5).

Email: brendanchambersbou@gmail.com (Brendan Chambers-Bourgeois)

Date of this document: 2026-06-05.

Table 1: SD-BKZ advantage on NTRU: tied $\rightarrow$ onset spike $\rightarrow$ plateau.

n	mean adv	std	seeds>0
59–67	~ 0	tight	3–9/20
71	+4.35	5.59	16/20
73	+9.68	5.21	19/20
79–89	+1.5	~ 0.2	20/20

- A cross-engine study (Section 6): G6K wired as a second engine behind a reproducibility-gated seam, a self-dual pump-and-jump construction validated against fplll's SD-BKZ, and confirmation that RHF-blindness is engine-independent but regime-dependent—consistent with independent observations of RHF-equality under a non-exact oracle [Row24].

- An independent re-validation of an fplll GSO catastrophic-cancellation fix on a new (NTRU) modulus family (Section 7).

This is a *technical report / data paper*: the emphasis is on the experimental apparatus, the data, and reproducibility, rather than on new theory. Closest prior work is Deng and Jia's self-dual BKZ over a sieve oracle [J+23]; we differ in measuring the SD-vs-BKZ structural difference via the reference-free profile observable rather than swapping the oracle for speed. White [Whi26] studies dynamic block-size strategies within the same Li–Nguyen framework, complementary to the fixed-β, q-swept regime characterised here.

2 Background

2.1 Circulant NTRU lattices and fatigue

2.2 BKZ, SD-BKZ, and the two engines

2.3 The reference-free DSD observable

3 Experimental Setup

3.1 Generators and engines

3.2 Reproducibility and determinism gates

3.3 Campaigns and parameter grids

4 The SD-BKZ Advantage Has a Dimension-Onset on NTRU

On circulant NTRU at $q = 97$ (standard regime), $\beta = 20$, 50 tours, the SD-BKZ-vs-BKZ advantage has a *sharp* dimension transition, not a smooth ramp (20 seeds per n):

Three regimes: tied for $n \leq 67$; an onset zone at $n = 71$–73 where the mean spikes to +4–10 nats with ∼25× the plateau variance (SD-BKZ cracks the dense sublattice on most-but-not-all instances); and a stable +1.5-nat plateau for $n \geq 79$ (std ≈0.2, 20/20 wins). The onset spike exceeds the plateau—non-monotonic.

Table 2: Reference-free DSD-onset: SD-BKZ reaches dense-sublattice discovery at lower q than BKZ, gap growing with n (∼15–20 seeds/cell).

n	SD onset q	BKZ onset q	gap%
67	146	149	2
79	175	175	0
89	237	281	18
101	426	514	21
113	732	932	27

Table 3: G6K SD-BKZ reaches DSD at lower q than BKZ ($n = 89$, $\beta = 40$, 20 seeds, $b_1 > 1.5$).

q	SD DSD	BKZ DSD	gap
97, 113	0/20	0/20	0 (pre-onset)
137	**14/20**	**4/20**	+10
157	20/20	19/20	+1
≥ 181	20/20	20/20	0 (saturated)

5 Fatigue and the Cross-Dimension DSD-Onset Gap

Sweeping q across the fatigue point at fixed n, the reference-free BKZ-vs-SD-BKZ profile divergence peaks near $q \approx q_{\mathrm{fat}}$. The SD-BKZ DSD-onset occurs at progressively *lower* q than the BKZ onset as n grows—the gap widens with dimension:

$n = 127$ is excluded (fplll §8 catastrophic cancellation + an off-grid crack).

6 Cross-Engine: SD-BKZ on a Sieve Oracle

RHF blindness is engine-independent but regime-dependent. On NTRU at $q = 97$, $\beta = 40$, the companion paper's thesis reproduces on the sieve: in the standard regime the RHF of BKZ and SD-BKZ bases is identical to machine precision while $d(\mathrm{LN})$ differs. The effect is regime-dependent—RHF-blindness breaks only on the rare seeds where SD-BKZ finds a short vector (which RHF reflects). This is consistent with RHF-equality reported under a non-exact (approximate-enumeration) oracle [Row24]; the present work measures the structure RHF misses.

Cross-engine validation of the construction. A dimension-resolved fplll-vs-G6K comparison ($n = 89/101/113$, 20 seeds each) confirms the self-dual G6K construction is sound: at $n = 101/113$ the two engines agree perfectly (no spurious sieve firing), and where a short vector genuinely exists ($n = 89$, seed 19) both engines land the identical first GS log-norm.

The DSD-onset gap reproduces on the sieve. A G6K DSD-onset q-sweep at $n = 89$, $\beta = 40$ shows that SD-BKZ reaches dense-sublattice discovery (reference-free: $b_1 > 1.5$, the q-independent ternary-secret floor $\approx \sqrt{2n \cdot 2/3}$) at *lower* q than BKZ.

At $q = 137$, SD-BKZ cracks DSD on 14/20 seeds where BKZ cracks only 4/20 (SD-only on 11): SD onset ≈137, BKZ onset ≈157. This is the DSD-onset gap of Table 2, now on a sieve oracle, at *lower* q than the $\beta = 20$ fplll onset (q=237)—stronger reduction lowers the onset. Crucially the gap is visible *only* reference-free: on the seeds where SD-BKZ cracks DSD, the *signed* $d(\mathrm{LN})$ advantage is large and *negative* (mean -30.8 at $q = 137$), because the DSD Z-shaped profile is far from the LWE-derived d(LN) reference. This makes the "signed d(LN) is a reference artifact on NTRU" point concrete and cross-engine: report reference-free DSD, never the signed advantage.

7 Re-Validating the fplll GSO Fix on a New Modulus Family

The companion paper documents a catastrophic-cancellation bug in fplll's squared-form GSO update at cryptographic moduli, and a Kahan-compensated fix. We re-validate that fix on a new (NTRU) modulus family: on $q \in \{971, 1087, 1201\}$ at $n = 127$, the four contaminated seeds recover from the precision-floor clamp ($b_1 = -345.388$) into the control band (≈ -0.1); the patched build passes all 15 fplll regression tests. The fix is thus validated on two modulus families (the companion paper's $q = 3329$ LWE, 38%→0% degeneracy, and this NTRU set).

8 Discussion

9 Reproducibility

10 Conclusion

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